

DBT task 1

- 1- mkdir dbt_project
- 2- cd dbt_project
- 3- Open VS Code from WSL using {code .}
- 4- VS Code communicates with WSL like a **small local server**, letting you edit Linux files from Windows.
- 5- Create a Python virtual environment write this command in terminal of vscode

```
python3 -m venv dbt-env
```

- Creates an **isolated Python environment** called dbt-env for this project.
- Keeps packages separate from system Python.

6- Activate the virtual environment

```
source dbt-env/bin/activate
```

7- Install dbt with Postgres adapter

```
pip install dbt-postgres
```

```
habiba@DESKTOP-M4F1VD9:~$ ls
Dockerfile  dbt_project  docker_kubernetes  etl.py  grafana_test  postgres-deployment.yaml
habiba@DESKTOP-M4F1VD9:~$ cd dbt_project
habiba@DESKTOP-M4F1VD9:~/dbt_project$ ls
dbt-env
habiba@DESKTOP-M4F1VD9:~/dbt_project$ |
```

8- Install dbt Core

```
pip install dbt-core
```

- 9- After that I tried to connect to any database in docker from postgres and after that create new database for the task called dbt_ecommerce_db

```
habiba@DESKTOP-M4F1VD9:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS                               NAMES
b894af3d757d   grafana/grafana:latest "/run.sh"              2 weeks ago   Up 25 seconds  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   grafana
b6b9ef0831c4   postgres:15           "docker-entrypoint.s..." 2 weeks ago   Up 25 seconds  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp   postgres

habiba@DESKTOP-M4F1VD9:~$ docker inspect postgres_db | grep -i env -A 5
"Env": [
  "POSTGRES_USER=admin",
  "POSTGRES_PASSWORD=admin123",
  "POSTGRES_DB=sales_db",
  "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/lib/postgresql/15/bin",
  "GOSU_VERSION=1.19",
]

habiba@DESKTOP-M4F1VD9:~$ psql -h localhost -p 5432 -U admin -d sales_db
Password for user admin:
psql: error: connection to server at "localhost" (127.0.0.1), port 5432 failed: FATAL: password authentication failed for user "admin"

habiba@DESKTOP-M4F1VD9:~$ psql -h localhost -p 5432 -U admin -d sales_db
Password for user admin:
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1), server 15.14 (Debian 15.14-1.pgdg13+1))
WARNING: psql major version 14, server major version 15.
Some psql features might not work.
Type "help" for help.

sales_db=# CREATE DATABASE dbt_ecommerce_db;
CREATE DATABASE
sales_db=# \c dbt_ecommerce_db
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1), server 15.14 (Debian 15.14-1.pgdg13+1))
WARNING: psql major version 14, server major version 15.
Some psql features might not work.
You are now connected to database "dbt_ecommerce_db" as user "admin".
dbt_ecommerce_db=# exit
```

- 1- docker start postgres_db
- 2- psql -h localhost -p 5432 -U admin -d sales_db
- 3- password is admin123
- 4- CREATE DATABASE dbt_ecommerce_db

```

"GOSU_VERSION=1.19",
habiba@DESKTOP-M4F1VD9:~$ psql -h localhost -p 5432 -U admin -d postgres
Password for user admin:
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1), server 15.14 (Debian 15.14-1.pgdg13+1))
WARNING: psql major version 14, server major version 15.
        Some psql features might not work.
Type "help" for help.

postgres=# help
You are using psql, the command-line interface to PostgreSQL.
Type:  \copyright for distribution terms
       \h for help with SQL commands
       \? for help with psql commands
       \g or terminate with semicolon to execute query
       \q to quit
postgres=# \l

```

List of databases					
Name	Owner	Encoding	Collate	Ctype	Access privileges
dbt_ecommerce_db	admin	UTF8	en_US.utf8	en_US.utf8	
postgres	admin	UTF8	en_US.utf8	en_US.utf8	
sales_db	admin	UTF8	en_US.utf8	en_US.utf8	
template0	admin	UTF8	en_US.utf8	en_US.utf8	=c/admin + admin=CTc/admin
template1	admin	UTF8	en_US.utf8	en_US.utf8	=c/admin + admin=CTc/admin

(5 rows)

10- make sure database is created after to connect in any database created before I list all database to make sure dbt_ecommerce created.

11- I have downloaded dbeaver and
connect to postgres container

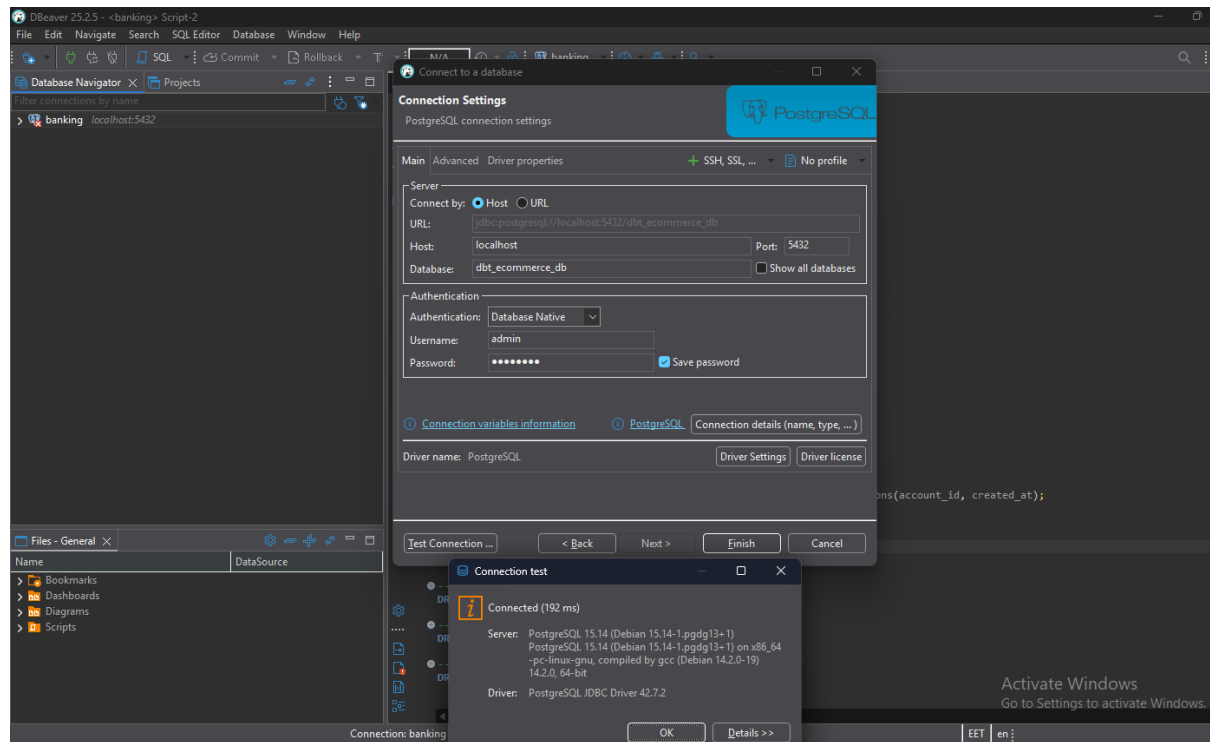
1- User admin

2- Password admin 123

3- Database name

dbt_ecommerce_db

4- Port 5432



12- Create two schema

```
dbt_ecommerce_db=# CREATE SCHEMA raw;
CREATE SCHEMA analytics;
CREATE SCHEMA
ERROR: schema "analytics" already exists
dbt_ecommerce_db=# \dn
List of schemas

```

Name	Owner
analytics	admin
public	pg_database_owner
raw	admin

```
(3 rows)
```

- 13- I make the dbt project using dbt init dbt_task and entered some requirements like in image and then the folder structure of dbt project appeared on left and update the dbt_project.yml So that it has **staging models** (to send them to the staging schema) **marts models** (to send them to the analytics schema)

```
habiba@DESKTOP-M4F1VD9:~/dbt_project$ source dbt-env/bin/activate
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project$ which dbt
/home/habiba/dbt_project/dbt-env/bin/dbt
```

```
• (dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project$ dbt init dbt_task
14:43:37 Running with dbt=1.10.15
14:43:37 Creating dbt configuration folder at /home/habiba/.dbt
14:43:37
Your new dbt project "dbt_task" was created!
```

For more information on how to configure the profiles.yml file,
please consult the dbt documentation here:

<https://docs.getdbt.com/docs/configure-your-profile>

One more thing:

Need help? Don't hesitate to reach out to us via GitHub issues or on Slack:

<https://community.getdbt.com/>

Happy modeling!

```
14:43:37 Setting up your profile.
Which database would you like to use?
[1] postgres
```

(Don't see the one you want? <https://docs.getdbt.com/docs/available-adapters>)

```
Enter a number: 1
host (hostname for the instance): localhost
port [5432]: 5432
user (dev username): admin
pass (dev password):
dbname (default database that dbt will build objects in): dbt_ecommerce_db
schema (default schema that dbt will build objects in): raw
threads (1 or more) [1]: 1
14:48:42 Profile dbt_task written to /home/habiba/.dbt/profiles.yml using target's profile_template.yml
your supplied values. Run 'dbt debug' to validate the connection.
```

DBT_PROJECT [WSL: UBUNTU-2...]

dbt_task

> analyses

> logs

> macros

> models

> seeds

> snapshots

> tests

◆ .gitignore

! dbt_project.yml

① README.md

> dbt-env

> logs

> OUTLINE

> TIMELINE

Start

Walkthroughs

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash - dbt_task

(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project\$ dbt debug

● (dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project\$ cd ~/dbt_project/dbt_task

● (dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task\$ dbt debug

14:51:43 Running with dbt=1.10.15

14:51:43 dbt version: 1.10.15

14:51:43 python version: 3.10.12

14:51:43 python path: /home/habiba/dbt_project/dbt-env/bin/python3

14:51:43 os info: Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.35

14:51:43 Using profiles dir at /home/habiba/.dbt

14:51:43 Using profiles.yml file at /home/habiba/.dbt/profiles.yml

14:51:43 Using dbt_project.yml file at /home/habiba/dbt_project/dbt_task/dbt_project.yml

14:51:43 adapter type: postgres

14:51:43 adapter version: 1.9.1

14:51:43 Configuration:

14:51:43 profiles.yml file [OK found and valid]

14:51:43 dbt_project.yml file [OK found and valid]

14:51:43 Required dependencies:

14:51:43 - git [OK found]

14:51:43 Connection:

14:51:43 host: localhost

14:51:43 port: 5432

14:51:43 user: admin

14:51:43 database: dbt_ecommerce_db

14:51:43 schema: raw

14:51:43 connect_timeout: 10

14:51:43 role: None

14:51:43 search_path: None

14:51:43 keepalives_idle: 0

14:51:43 sslmode: None

14:51:43 sslcert: None

14:51:43 sslkey: None

14:51:43 sslrootcert: None

14:51:43 application_name: dbt

14:51:43 retries: 1

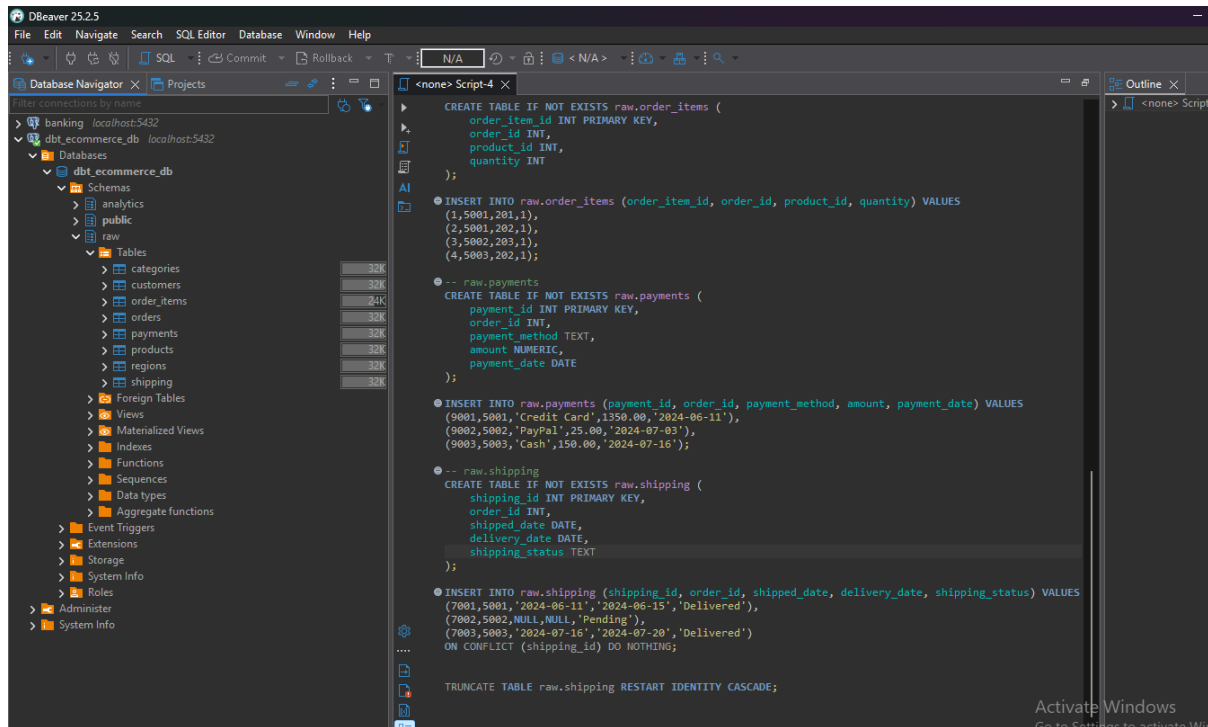
14:51:43 Registered adapter: postgres=1.9.1

14:51:43 Connection test: [OK connection ok]

14:51:43 All checks passed!

(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task\$

14- Create raw data



15- Then i want to run my models so i created a schema called staging in my db then update profile.yml and set schema from raw to staging and then ran my models using dbt run so it ran and made 8 views in staging schema

```
{ dbt run --select staging }
```

```
dbt_task > models > staging > ! sources.yml
3 sources:
10 - name: products
11 - name: orders

(dbt-env) habiba@DESKTOP-M4F1V09:~/dbt
(dbt-env) habiba@DESKTOP-M4F1V09:~/dbt
(dbt-env) habiba@DESKTOP-M4F1V09:~/dbt_project/dbt_task$ dbt clean
dbt run --select staging
19:06:05 Running with dbt=1.10.15
19:06:05 Registered adapter: postgres=1.9.1
19:06:05 Unable to do partial parsing because saved manifest not found. Starting full parse.
19:06:06 [WARNING]: Configuration paths exist in your dbt_project.yml file which do not apply to any resources.
There are 1 unused configuration paths:
- models.dbt_ecommerce.marts
19:06:06 Found 10 models, 4 data tests, 8 sources, 459 macros
19:06:06 Concurrency: 1 threads (target='dev')
19:06:06
19:06:06 1 of 8 START sql view model raw_staging.stg_categories ..... [RUN]
19:06:07 1 of 8 OK created sql view model raw_staging.stg_categories ..... [CREATE VIEW in 0.08s]
19:06:07 2 of 8 START sql view model raw_staging.stg_customers ..... [RUN]
19:06:07 2 of 8 OK created sql view model raw_staging.stg_customers ..... [CREATE VIEW in 0.03s]
19:06:07 3 of 8 START sql view model raw_staging.stg_order_items ..... [RUN]
19:06:07 3 of 8 OK created sql view model raw_staging.stg_order_items ..... [CREATE VIEW in 0.03s]
19:06:07 4 of 8 START sql view model raw_staging.stg_orders ..... [RUN]
19:06:07 4 of 8 OK created sql view model raw_staging.stg_orders ..... [CREATE VIEW in 0.03s]
19:06:07 5 of 8 START sql view model raw_staging.stg_payments ..... [RUN]
19:06:07 5 of 8 OK created sql view model raw_staging.stg_payments ..... [CREATE VIEW in 0.03s]
19:06:07 6 of 8 START sql view model raw_staging.stg_products ..... [RUN]
19:06:07 6 of 8 OK created sql view model raw_staging.stg_products ..... [CREATE VIEW in 0.03s]
19:06:07 7 of 8 START sql view model raw_staging.stg_regions ..... [RUN]
19:06:07 7 of 8 OK created sql view model raw_staging.stg_regions ..... [CREATE VIEW in 0.03s]
19:06:07 8 of 8 START sql view model raw_staging.stg_shipping ..... [RUN]
19:06:07 8 of 8 OK created sql view model raw_staging.stg_shipping ..... [CREATE VIEW in 0.03s]
19:06:07
19:06:07 Finished running 8 view models in 0 hours 0 minutes and 0.42 seconds (0.42s).
19:06:07
19:06:07 Completed successfully
19:06:07
19:06:07 Done. PASS=8 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=8
(dbt-env) habiba@DESKTOP-M4F1V09:~/dbt_project/dbt_task$
```

16- then i made aggregations by making directory marts inside models and made 3 files that will contain my aggregation models

File Edit Selection View Go Run ... dbt_project [WSL: Ubuntu-22.04]

EXPLORER

- DBT_PROJE...
- dbt_task
 - analyses
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 - marts
 - agg_customer_spendi...
 - agg_product_performa...
 - agg_sales.sql
 - staging
 - seeds
 - snapshots
 - target
 - tests
 - .gitignore
 - dbt_project.yml
 - README.md
 - dbt-env
 - logs
- OUTLINE

dbt_task > models > marts > agg_product_performance.sql

```
1 select
2   p.product_id,
3   p.product_name,
4   c.category_name,
5   sum(oi.quantity) as total_quantity_sold,
6   sum(oi.quantity * p.price) as total_sales
7 from {{ ref('stg_products') }} p
8 left join {{ ref('stg_categories') }} c
9   on p.category_id = c.category_id
10 left join {{ ref('stg_order_items') }} oi
11   on p.product_id = oi.product_id
12 group by 1,2,3
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task\$ dbt run --select marts

19:26:36 Running with dbt=1.10.15

19:26:36 Registered adapter: postgres=1.9.1

19:26:36 Found 13 models, 4 data tests, 8 sources, 459 macros

19:26:36 Concurrency: 1 threads (target='dev')

19:26:36 1 of 3 START sql table model raw_analytics.agg_customer_spending [RUN]

19:26:37 1 of 3 OK created sql table model raw_analytics.agg_customer_spending [SELECT 3 in 0.09s]

19:26:37 2 of 3 START sql table model raw_analytics.agg_product_performance [RUN]

19:26:37 2 of 3 OK created sql table model raw_analytics.agg_product_performance [SELECT 3 in 0.04s]

19:26:37 3 of 3 START sql table model raw_analytics.agg_sales [RUN]

19:26:37 3 of 3 OK created sql table model raw_analytics.agg_sales [SELECT 3 in 0.03s]

19:26:37 Finished running 3 table models in 0 hours 0 minutes and 0.31 seconds (0.31s).

19:26:37 Completed successfully

19:26:37 Done. PASS=3 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=3

(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task\$

File Edit Selection View Go Run ... dbt_project [WSL: Ubuntu-22.04]

EXPLORER

- DBT_PROJE...
- dbt_task
 - analyses
 - logs
 - macros
 - models
 - example
 - marts
 - agg_customer_spendi...
 - agg_product_performa...
 - agg_sales.sql
 - staging
 - seeds
 - snapshots
 - target
 - tests
 - .gitignore
 - dbt_project.yml
 - README.md
 - dbt-env
 - logs

dbt_task > models > marts > agg_customer_spending.sql

```
1 select
2
3   c.first_name,
4   c.last_name,
5   r.region_name,
6   count(o.order_id) as total_orders,
7   sum(o.total_amount) as total_spent
8 from {{ ref('stg_customers') }} c
9 left join {{ ref('stg_regions') }} r
10  on c.region_id = r.region_id
11 left join {{ ref('stg_orders') }} o
12  on c.customer_id = o.customer_id
13 group by 1,2,3,4
14
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash - dbt_task

```
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$ dbt run --select marts
19:26:36 Running with dbt=1.10.15
19:26:36 Registered adapter: postgres=1.9.1
19:26:36 Found 13 models, 4 data tests, 8 sources, 459 macros
19:26:36 Concurrency: 1 threads (target='dev')
19:26:36
19:26:36 1 of 3 START sql table model raw_analytics.agg_customer_spending ..... [RUN]
19:26:37 1 of 3 OK created sql table model raw_analytics.agg_customer_spending ..... [SELECT 3 in 0.09s]
19:26:37 2 of 3 START sql table model raw_analytics.agg_product_performance ..... [RUN]
19:26:37 2 of 3 OK created sql table model raw_analytics.agg_product_performance ..... [SELECT 3 in 0.04s]
19:26:37 3 of 3 START sql table model raw_analytics.agg_sales ..... [RUN]
19:26:37 3 of 3 OK created sql table model raw_analytics.agg_sales ..... [SELECT 3 in 0.03s]
19:26:37
19:26:37 Finished running 3 table models in 0 hours 0 minutes and 0.31 seconds (0.31s).
19:26:37
19:26:37 Completed successfully
19:26:37
19:26:37 Done. PASS=3 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=3
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$
```

Activate Windows
Go to Settings to activate Windows.

```
dbt_task > models > marts > agg_sales.sql
1  select
9    sum(p.price * oi.quantity) as calculated_amount
10   from {{ ref('stg_orders') }} o
11   left join {{ ref('stg_order_items') }} oi
12     on o.order_id = oi.order_id
13   left join {{ ref('stg_products') }} p
14     on oi.product_id = p.product_id
15   left join {{ ref('stg_customers') }} c
16     on o.customer_id = c.customer_id
17   left join {{ ref('stg_regions') }} r
18     on c.region_id = r.region_id
19   group by 1,2,3,4,5,6
20
```

```
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$ dbt run --select marts
19:26:36 Running with dbt=1.10.15
19:26:36 Registered adapter: postgres=1.9.1
19:26:36 Found 13 models, 4 data tests, 8 sources, 459 macros
19:26:36 Concurrency: 1 threads (target='dev')
19:26:36
19:26:36 1 of 3 START sql table model raw_analytics.agg_customer_spending ..... [RUN]
19:26:37 1 of 3 OK created sql table model raw_analytics.agg_customer_spending ..... [SELECT 3 in 0.09s]
19:26:37 2 of 3 START sql table model raw_analytics.agg_product_performance ..... [RUN]
19:26:37 2 of 3 OK created sql table model raw_analytics.agg_product_performance ..... [SELECT 3 in 0.04s]
19:26:37 3 of 3 START sql table model raw_analytics.agg_sales ..... [RUN]
19:26:37 3 of 3 OK created sql table model raw_analytics.agg_sales ..... [SELECT 3 in 0.03s]
19:26:37
19:26:37 Finished running 3 table models in 0 hours 0 minutes and 0.31 seconds (0.31s).
19:26:37 Completed successfully
19:26:37 Done. PASS=3 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=3
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$
```

17- I made tests by making 2 files inside test directory , stg_orders_test.yml and stg_payment_test.yml and wrote test content and ran dbt test to run all tests if i want some test only just add --select and name of test file and they all passes the test.

Visual Studio Code interface showing a dbt project in WSL (Ubuntu-22.04). The Explorer pane on the left shows the project structure, including files like `sources.yml`, `agg_sales.sql`, `agg_customer_spending.sql`, `agg_product_performance.sql`, and `stg_orders_test.yml`. The main editor displays the content of `stg_orders_test.yml`:

```
1 version: 2
2
3 models:
4   - name: stg_orders
5     columns:
6       - name: order_id
7         tests:
8           - not_null
9           - unique
10
```

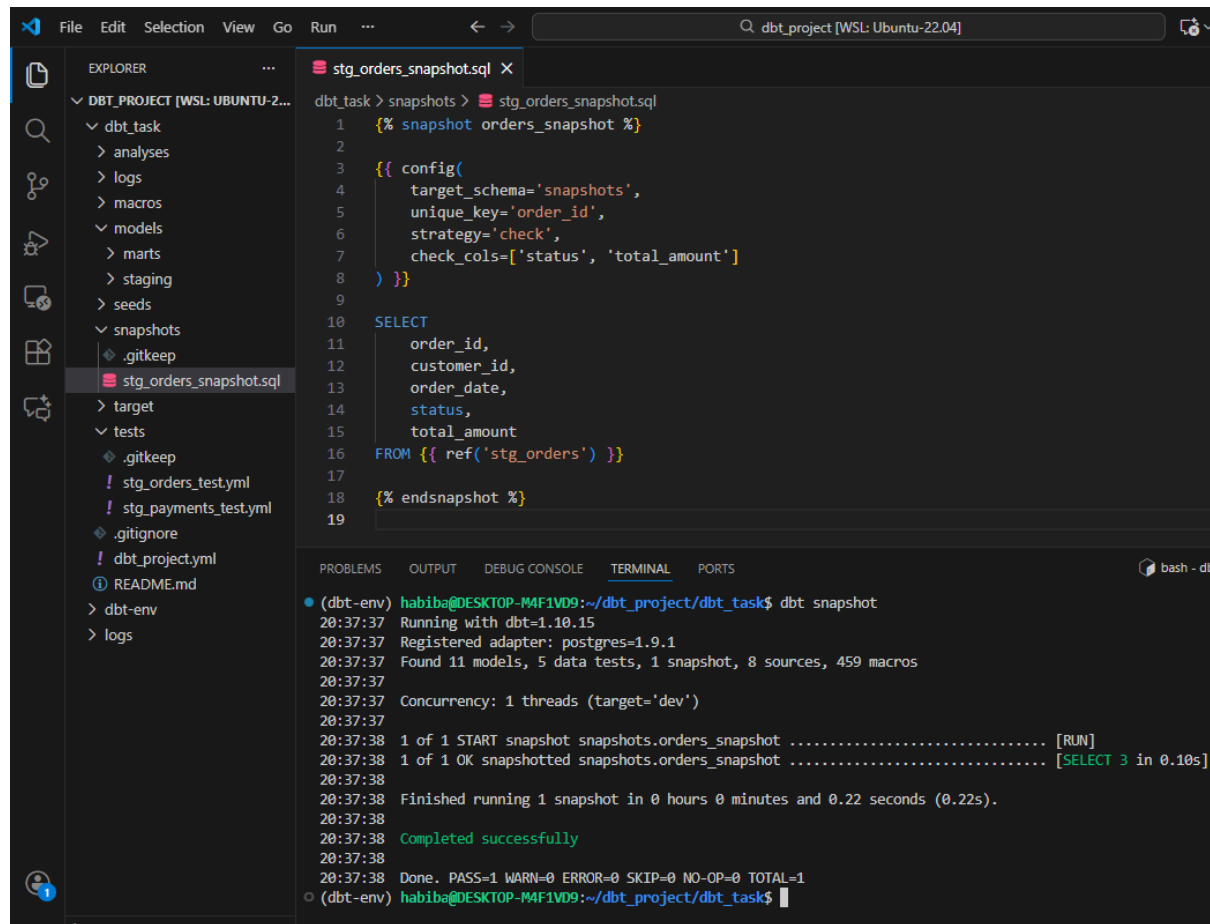
The TERMINAL pane at the bottom shows the output of the `dbt test` command:

```
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$ dbt test
20:25:11 Running with dbt=1.10.15
20:25:11 Registered adapter: postgres=1.9.1
20:25:12 Found 11 models, 5 data tests, 8 sources, 459 macros
20:25:12
20:25:12 Concurrency: 1 threads (target='dev')
20:25:12
20:25:12 1 of 5 START test not_null_stg_orders_order_id ..... [RUN]
20:25:12 1 of 5 PASS not_null_stg_orders_order_id ..... [PASS in 0.04s]
20:25:12 2 of 5 START test not_null_stg_payments_amount ..... [RUN]
20:25:12 2 of 5 PASS not_null_stg_payments_amount ..... [PASS in 0.02s]
20:25:12 3 of 5 START test not_null_stg_payments_payment_id ..... [RUN]
20:25:12 3 of 5 PASS not_null_stg_payments_payment_id ..... [PASS in 0.02s]
20:25:12 4 of 5 START test unique_stg_orders_order_id ..... [RUN]
20:25:12 4 of 5 PASS unique_stg_orders_order_id ..... [PASS in 0.02s]
20:25:12 5 of 5 START test unique_stg_payments_payment_id ..... [RUN]
20:25:12 5 of 5 PASS unique_stg_payments_payment_id ..... [PASS in 0.02s]
20:25:12
20:25:12 Finished running 5 data tests in 0 hours 0 minutes and 0.22 seconds (0.22s).
20:25:12
20:25:12 Completed successfully
20:25:12
20:25:12 Done. PASS=5 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=5
(dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$
```

```
dbt_task > tests > ! stg_payments_test.yml
1 version: 2
2
3 models:
4   - name: stg_payments
5     columns:
6       - name: payment_id
7         tests:
8           - not_null
9           - unique
10
11     - name: amount
12       tests:
13         - not_null
14
```

```
bash - dbt_task
• (dbt-env) habiba@DESKTOP-M4F1V09:~/dbt_project/dbt_task$ dbt test
20:25:11 Running with dbt=1.10.15
20:25:11 Registered adapter: postgres=1.9.1
20:25:12 Found 11 models, 5 data tests, 8 sources, 459 macros
20:25:12 Concurrency: 1 threads (target='dev')
20:25:12
20:25:12 1 of 5 START test not_null_stg_orders_order_id ..... [RUN]
20:25:12 1 of 5 PASS not_null_stg_orders_order_id ..... [PASS in 0.04s]
20:25:12 2 of 5 START test not_null_stg_payments_amount ..... [RUN]
20:25:12 2 of 5 PASS not_null_stg_payments_amount ..... [PASS in 0.02s]
20:25:12 3 of 5 START test not_null_stg_payments_payment_id ..... [RUN]
20:25:12 3 of 5 PASS not_null_stg_payments_payment_id ..... [PASS in 0.02s]
20:25:12 4 of 5 START test unique_stg_orders_order_id ..... [RUN]
20:25:12 4 of 5 PASS unique_stg_orders_order_id ..... [PASS in 0.02s]
20:25:12 5 of 5 START test unique_stg_payments_payment_id ..... [RUN]
20:25:12 5 of 5 PASS unique_stg_payments_payment_id ..... [PASS in 0.02s]
20:25:12
20:25:12 Finished running 5 data tests in 0 hours 0 minutes and 0.22 seconds (0.22s).
20:25:12
20:25:12 Completed successfully
20:25:12
20:25:12 Done. PASS=5 WARN=0 ERROR=0 SKIP=0 NO_OP=0 TOTAL=5
```

18- I made snapshots by first creating a schema called snapshots in db and in dbt project inside snapshot directory i made a file called stg_orders._snapshot.sql and wrote snapshot content



```
File Edit Selection View Go Run ... dbt_project [WSL: Ubuntu-22.04]
```

```
EXPLORER
DBT_PROJECT [WSL: UBUNTU-22.04]
  dbt_task
    analyses
    logs
    macros
    models
    marts
    staging
    seeds
    snapshots
      stg_orders_snapshot.sql
      target
    tests
      .gitkeep
      stg_orders_test.yml
      stg_payments_test.yml
      .gitignore
      dbt_project.yml
      README.md
    dbt-env
    logs
```

```
stg_orders_snapshot.sql
1  {% snapshot orders_snapshot %}
2
3  {{ config(
4    target_schema='snapshots',
5    unique_key='order_id',
6    strategy='check',
7    check_cols=['status', 'total_amount']
8  ) }}
9
10 SELECT
11   order_id,
12   customer_id,
13   order_date,
14   status,
15   total_amount
16 FROM {{ ref('stg_orders') }}
17
18 {% endsnapshot %}
19
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
• (dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$ dbt snapshot
20:37:37 Running with dbt=1.10.15
20:37:37 Registered adapter: postgres=1.9.1
20:37:37 Found 11 models, 5 data tests, 1 snapshot, 8 sources, 459 macros
20:37:37 Concurrency: 1 threads (target='dev')
20:37:37
20:37:38 1 of 1 START snapshot snapshots.orders_snapshot ..... [RUN]
20:37:38 1 of 1 OK snapshot snapshots.orders_snapshot ..... [SELECT 3 in 0.10s]
20:37:38
20:37:38 Finished running 1 snapshot in 0 hours 0 minutes and 0.22 seconds (0.22s).
20:37:38 Completed successfully
20:37:38 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=1
○ (dbt-env) habiba@DESKTOP-M4F1VD9:~/dbt_project/dbt_task$
```

- 19- To make sure snapshot work correct I have print value of id 5001 after that make ubdate for this value in raw table to haboba and run after that
 - 1- dbt run -m stg_orders
 - 2- dbt snapshot
- 20- then I return to print the row again of id 5001 it print to me the two values, the old one and the new one , so this mean work correctly

DBeaver 25.2.5 - <dbt_ecommerce_db> Script-5

File Edit Navigate Search SQL Editor Database Window Help

Auto dbt_ecommerce_db public@dbt_ecommerce_db

Database Navigator Projects

Filter connections by name

- banking localhost:5432
- dbt_ecommerce_db localhost:5432
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 - dbt_ecommerce_db
 - Schemas
 - analytics
 - public
 - raw
 - raw_analytics
 - raw_staging
 - Tables
 - Foreign Tables
 - Views
 - Materialized Views
 - Indexes
 - Functions
 - Sequences
 - Data types
 - Aggregate functions
 - snapshots

SQL Editor

```
SELECT *
FROM snapshots.orders_snapshot
WHERE order_id = 5001
ORDER BY dbt_valid_from;
```

orders_snapshot 1

Enter a SQL expression to filter results (use Ctrl+Space)

id	order_id	customer_id	order_date	AZ status	total_amount	AZ
1	5,001	1	2024-06-10	Shipped	1,350	abf

Statistics 1

Name	Value
Updated Rows	1
Execute time	0.0s
Start time	Fri Nov 21 22:52:45 EET 2025
Finish time	Fri Nov 21 22:52:45 EET 2025
Query	UPDATE raw.orders SET status = 'haboba' WHERE order_id = 5001

DBeaver 25.2.5 - <dbt_ecommerce_db> Script-5

File Edit Navigate Search SQL Editor Database Window Help

Auto dbt_ecommerce_db public@dbt_ecommerce_db

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 - Sequences
 - Data types
 - Aggregate functions
 - snapshots
 - Tables
 - orders_snapshot 16K
 - Columns
 - 123 order_id (int4)

SQL Editor

```
UPDATE raw.orders
SET status = 'haboba'
WHERE order_id = 5001;
```

Statistics 1

Name	Value
Updated Rows	1
Execute time	0.0s
Start time	Fri Nov 21 22:52:45 EET 2025
Finish time	Fri Nov 21 22:52:45 EET 2025
Query	UPDATE raw.orders SET status = 'haboba' WHERE order_id = 5001

DBBeaver 25.2.5 - <dbt_ecommerce_db> Script-5

File Edit Navigate Search SQL Editor Database Window Help

Auto dbt_ecommerce_db public@dbt_ecommerce_db

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```
SELECT *
FROM snapshots.orders_snapshot
WHERE order_id = 5001
ORDER BY dbt_valid_from;
```

orders_snapshot 1

Enter a SQL expression to filter results (use Ctrl+Space)

	order_id	customer_id	order_date	status	total_amount	AZ
1	5,001	1	2024-06-10	Shipped	1,350	abf
2	5,001	1	2024-06-10	haboba	1,350	d69
						5001